Structured questions

|!|BCELB0040000001|!| ★ **SI**

Water may move into or out of potato tissue by osmosis.

The list below shows some apparatus that could be used to investigate how the concentration of sucrose solution affects this movement of water.

electronic balance	beakers	boiling tubes
test tube rack	measuring cylinders	sucrose solution (20%)
razor blade	white tile	blotting paper
cork borer	rubber stoppers	distilled water
potato tuber	ruler	

(a) What is osmosis? (2 marks)

(b) Describe how you would use the apparatus in the list to investigate the movement of water into and out of potato tissue. You may also use other apparatus if you wish. (8 marks)

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(c) What results would you expect to obtain? (2 marks)

_____

_____

##

(a) Osmosis refers to the net movement of water molecules from a region of higher water potential to a region of lower water potential (1)
across a differentially permeable membrane. (1)

(b) By mixing with known volumes of distilled water, dilute the 20% sucrose solution into a series of sucrose solutions of different concentrations. (1)
Use the cork borer to obtain five strips from the potato tuber. Cut each strip to 5 cm long. (1)
Gently blot the potato strips with blotting paper. (1)
Measure the initial mass of each potato strip with an electronic balance. (1)
Add a potato strip to each of the five boiling tubes, each containing a different concentration of sucrose solution (20%, 15%, 10%, 5%, 0%). (1)
Cover the boiling tubes with rubber stoppers and leave them for one hour. (1)
Remove the strips and blot them with blotting paper. (1)
Measure and record the final mass of each strip. (1)

(c) The mass of potato strips immersed in dilute sucrose solution increases. (1)
The mass of potato strips immersed in concentrated sucrose solution decreases. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Practical skills	8
(c)	Applying concepts	2

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|!|BCELA0040000002|!| ★★

Vivian performed an experiment to study osmosis using potato tubers. She obtained five potato strips and measured the mass of each strip with an electronic balance. She then immersed each strip in a different concentration of sucrose solution. The table below shows the results.

Concentration of sucrose solution (mol / dm^3)	**Mass of potato strip**		**Percentage change in mass**
	initial mass	**final mass**	
0.0	5.00	5.50	10%
0.2	4.90	5.12	4.5%
0.4	5.10	5.12	0.4%
0.6	4.90	4.75	-3.0%
0.8	5.10	4.80	x

(a) Calculate the percentage change in mass of the potato strip immersed in 0.8 mol / dm^3 sucrose solution. Show your working. (2 marks)

(b) On the graph paper provided, plot a graph to show the results of Vivian's experiment. (4 marks)

(c) Describe what happened to the potato strips immersed in sucrose solution with a concentration less than 0.4 mol / dm^3. Explain your answer. (3 marks)

(d) A potato strip has been immersed in a sucrose solution of unknown concentration. Its mass changed from 5.0 g to 5.15 g.
Using the graph you have plotted, find the concentration of this sucrose solution. (2 marks)

##

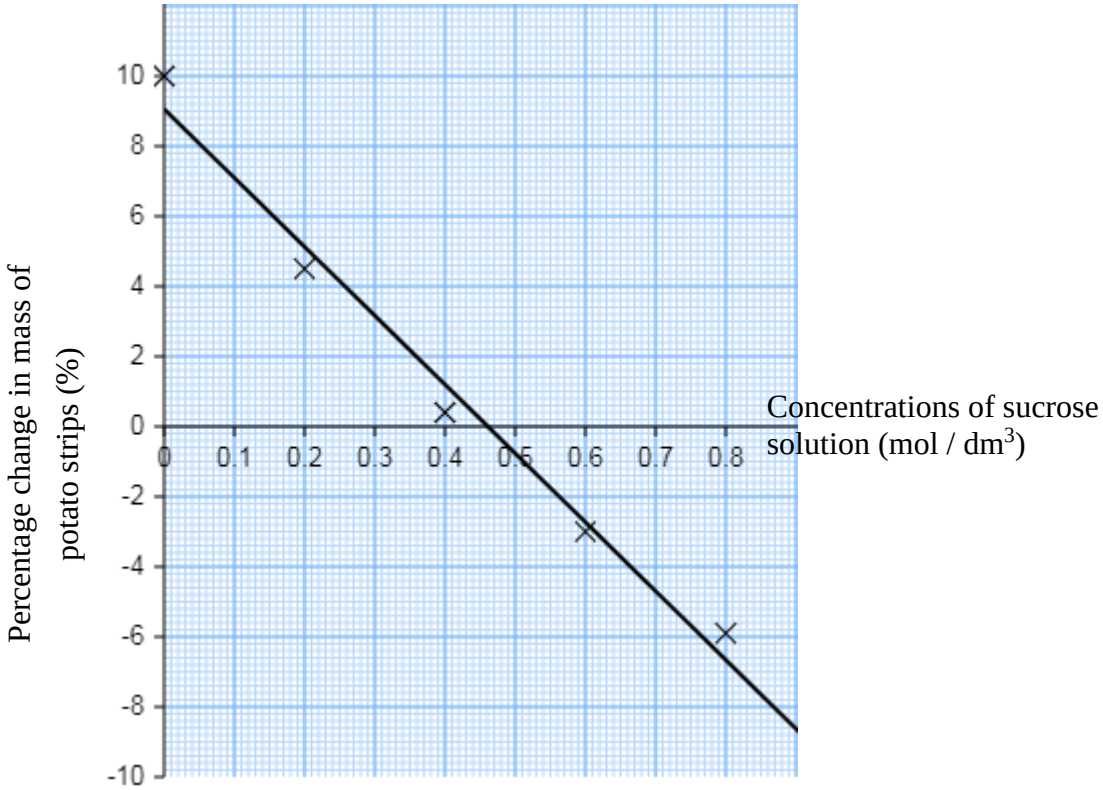
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(a) (4.8–5.1) / 5.1 × 100% (1)

= –5.9% (1)

(b)

Percentage change in mass of potato strips immersed in different concentrations of sucrose solution



X- and Y-axes with correct names, units and scales (2)

Correct plotting and joining of the five points (1)

Correct title (1)

(c) The water potential of the sucrose solution with a concentration less than 0.4 mol / dm^3 is higher than that inside the potato cells. (1)

There is a net gain of water by osmosis. (1)

The mass of the potato strips increases. (1)

(d) (5.15–5.00) / 5.00 × 100% = 3% (1)

From the graph, the concentration of sucrose solution which produces a 3% increase in mass is 0.3 mol / dm^3. (1)

(11 marks)

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Presenting information	4
(c)	Analysing data	1
	Applying concepts	2
(d)	Mathematical skills	1
	Analysing data	1

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|!|BCELA0040000003|!| ★★ **SI**

James compared the water potential of four liquids commonly found in the kitchen using naked eggs (eggs with their outer shell removed). The naked eggs were immersed into the liquids for five hours. The table below shows the results.

Mass of an naked egg	**Solution**			
	Grape juice	**Soy sauce**	**Corn syrup**	**Vinegar**
Before immersion	95 g	92 g	94 g	93 g
After immersion	61 g	46 g	50 g	125 g
Percentage change in mass	–35.8%		–46.8%	

(a) What should James do before weighting the naked eggs? Explain the significance of this step. (2 marks)

(b) (i) Explain why percentage change in the mass of eggs, rather than change in mass, is used to compare the water potential of the liquids. (1 mark)

(ii) Calculate the percentage change in the mass of eggs in soy sauce and vinegar, and then arrange the water potential of the four liquids in ascending order. (3 marks)

(c) Design an experiment to determine the concentration of corn syrup that has the same water potential as an naked egg. (4 marks)

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##

(a) Dry the surfaces of the eggs with a paper towel. (1)

To make sure that the change in mass measured is due to the egg content only. (1)

(b) (i) The mass of eggs are different at the start of experiment. Comparing the percentage change in mass cancels out the initial differences in the mass of the eggs. (1)

(ii) Percentage change in the mass of egg in soy sauce
= (46–92) / 92 × 100% = –50% (1)

Percentage change in the mass of egg in vinegar
= (125–93) / 93 × 100% = +34.4% (1)

The water potential of the liquids in ascending order:
Soy sauce < Corn syrup < Grape juice < Vinegar (1)

(c) Prepare a series of corn syrup solutions of different concentrations (e.g. without dilution, 1:1, 1:2, 1:5 and 1:10). (1)

Immerse naked eggs into the corn syrup solutions of different concentration and measure the change in the mass of naked eggs after 24 hours. (1)

Plot a graph of the percentage change in the mass of naked eggs against the concentration of corn syrup. (1)

The point at which the curve intercepts the X-axis represents that there is no net gain or loss of water by osmosis by the eggs at that concentration. (1)

This concentration of corn syrup has the same water potential as the naked eggs.

(10 marks)

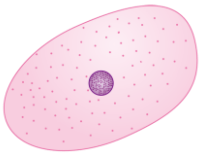
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	Skills assessed	Marks
(a)	Practical skills	2
(b)(i)	Practical skills	1
(b)(ii)	Mathematical skills	2
	Analysing data	1
(c)	Practical skills	4

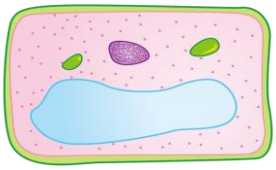
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|!|BCELC0040000004|!|

The diagrams below show the appearance of an animal cell and a plant cell observed under a light microscope.

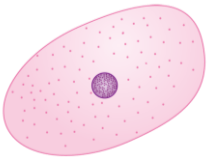
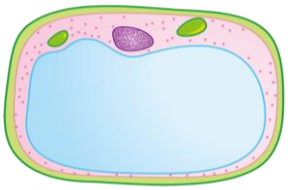
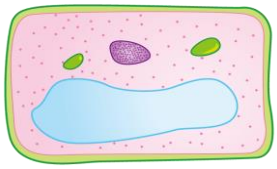
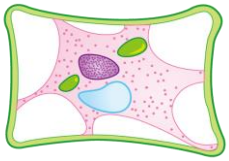


Animal cell



Plant cell

To study osmosis, the animal cells and plant cells were placed in solutions of different salt concentrations. After 20 minutes, a sample of cells was taken from each solution and examined with the microscope. The observations are shown as follows.

Animal cell	**Plant cell**
1. 	4. 
2. No intact cells can be seen.	5. 
3. 	6. 

(a) What is osmosis? (2 marks)

_____

_____

(b) Which animal cell had been placed in a hypertonic solution? (1 mark)

_____

(c) Use a suitable term to describe the condition of cell 6. (1 mark)

_____

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(d) In conditions 2 and 4, the cells had been placed in solutions of the same concentration. Describe and explain the different results observed. (4 marks)

##

(a) Osmosis refers to the net movement of water molecules from a region of higher water potential to a region of lower water potential (1)
across a differentially permeable membrane. (1)
(b) cell 3 (1)
(c) plasmolysis (1)
(d) Animal cells do not have a cell wall. (1)
They swell and then burst. (1)
Plant cells have a cell wall which prevents the cell from bursting. (1)
They swell and become turgid. (1)

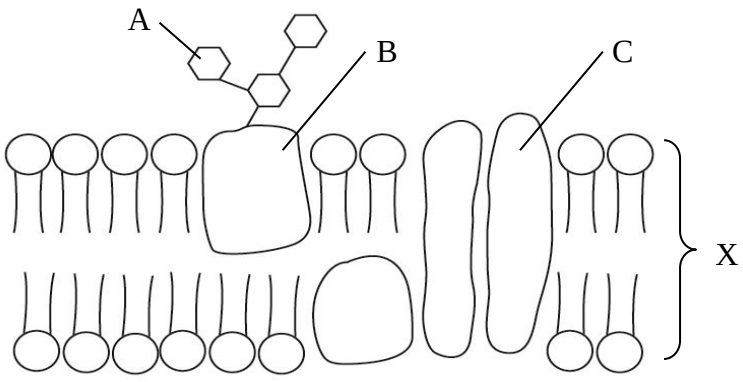
(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Analysing data	1
(c)	Understanding basic concepts	1
(d)	Analysing data	2
	Applying concepts	2

|!|BCELC0040000005|!|

The diagram below shows a model of the cell membrane.



(a) Name this model. Give two features to explain why such name is given to this model. (3 marks)

(b) Name the molecule formed by A and B. Give ***one*** function of this molecule. (2 marks)

(c) Name structure C. Suggest ***one*** substance which can pass through it. (2 marks)

(d) Suggest ***two*** substances which can pass layer X. (2 marks)

##

(a) fluid mosaic model (1)
The cell membrane is 'fluid' in nature: the phospholipid molecules and protein molecules can move laterally within the bilayer. (1)
Mosaic: protein molecules intersperse within the phospholipid bilayer. (1)
(b) A and B form the glycoprotein. (1)
It acts as an recognition marker. (1)
(c) C is a channel protein. (1)
Water / ions can pass through C. (1)
(d) Oxygen / carbon dioxide / lipid-soluble substance / other reasonable answers (1, 1)

(9 marks)

##

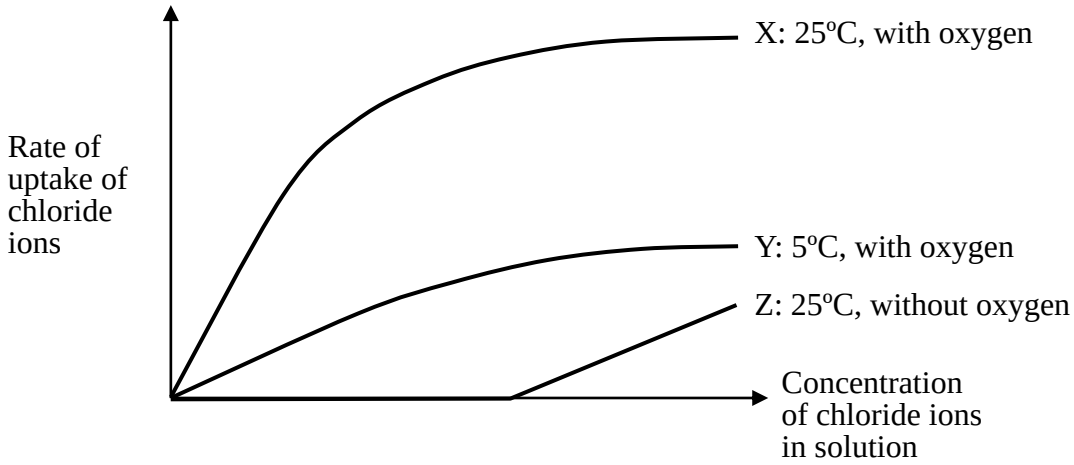
	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Analysing data	1
	Understanding basic concepts	1

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(c)	Analysing data	1
	Understanding basic concepts	1
(d)	Applying concepts	2

|!|BCELA0040000006|!| ★★

In an experiment, plants were grown with their roots immersed in solutions of different conditions. The rate of uptake of chloride ions by the plant roots was measured during the experiment. The results are shown below.



(a) Suggest the process by which the root cells absorb chloride ions. Support your answer with evidence from the graph. (2 marks)

(b) According to the graph, state ***two*** factors that affect the rate of uptake of chloride ions by the roots. (2 marks)

(c) Describe and explain the shape of curve X. (2 marks)

(d) Explain for the difference in the rate of chloride uptake between 5 ºC and 25 ºC in the presence of oxygen. (2 marks)

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(e) Describe and explain the shape of curve Z. (4 marks)

##

(a) Process: active transport (1)
Evidence:
It occurs when oxygen is present (requires respiration). OR
It occurs against a concentration gradient. (1)

(b) Any ***two*** of the following: (2)
- concentration of chloride ions in solution
- temperature
- oxygen availability

(c) Curve X first rises and finally levels off / reaches a plateau (1)
because the carrier proteins used in active transport are saturated with chloride ions. (1)

(d) Any ***one set*** of the following: (2)
- At a higher temperature, the rate of respiration increases (1)
  that leads to increased ATP production / energy release for active transport. (1)
- At a higher temperature, the chance of collision between chloride ions and carrier proteins in the cell membrane of root cell increases (1)
  because of higher kinetic energy. (1)

(e) When the chloride concentration is low (lower than that in the cytoplasm of root cells), there is no uptake of chloride ions by root cells. (1)
This is because there is no net uptake of chloride ions by diffusion OR
the absence of oxygen cannot support the uptake of chloride ions by active transport. (1)
When the chloride concentration is high (higher than that in the cytoplasm of root cells), the uptake of chloride ions increases steadily. (1)
This is because the diffusion rate of chloride ions is proportional to the concentration gradient. (1)

(12 marks)

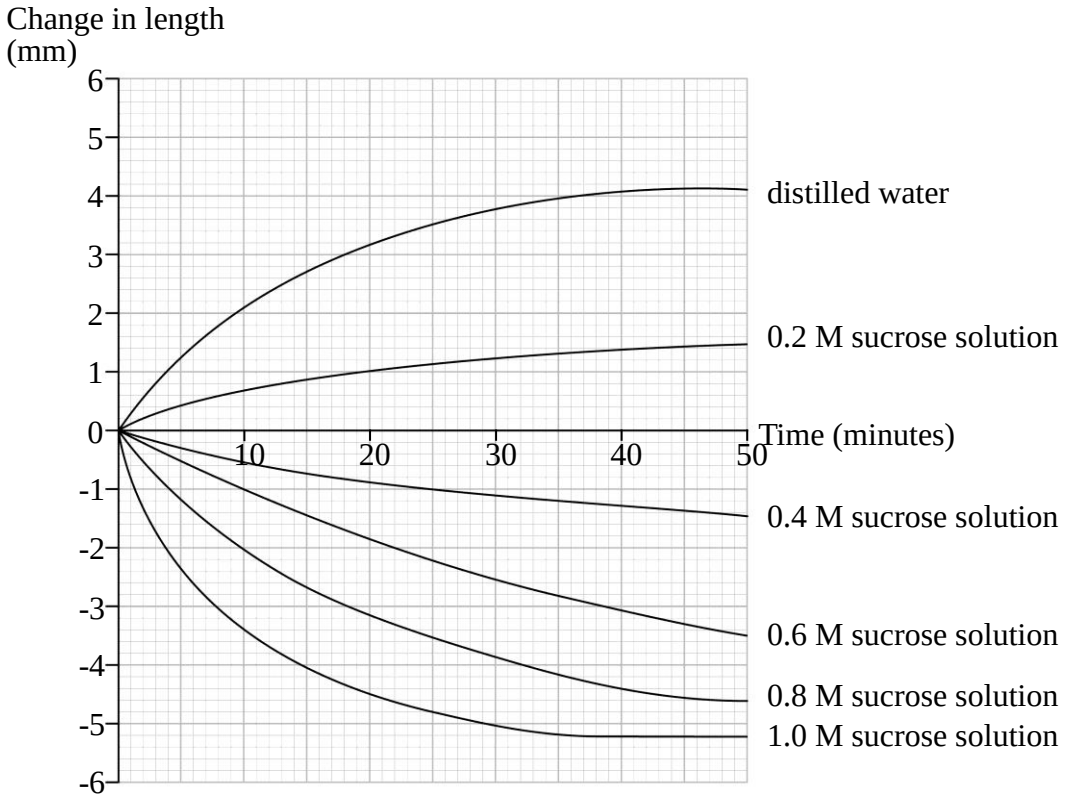
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##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	2
(c)	Analysing data	1
	Applying concepts	1
(d)	Applying concepts	2
(e)	Analysing data	2
	Applying concepts	2

|!|BCELB0040000007|!| ★

In an experiment, six cylinders of the same size and mass were cut from a potato. One of them was placed in distilled water and the others were placed in sucrose solutions of different concentrations. The lengths of the potato cylinders were measured every five minutes. The following graph shows the changes in the length of the cylinders.



(a) Explain the change in the length of the potato cylinder in distilled water. (4 marks)

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(b) The length of the potato cylinder in 1.0 M sucrose solution stopped changing after 40 minutes. Explain why. (2 marks)

(c) Explain the difference in the rate of decrease in the length between the potato cylinders in 0.4 M and 0.6 M sucrose solutions. (4 marks)

(d) Suggest ***one*** importance of osmosis to young seedlings. (1 mark)

##

(a) The length of the potato cylinder in distilled water increases. (1)
This is because the water potential of distilled water is higher than that of the potato cylinder. (1)
There is a net movement of water molecules into the potato cells. (1)
The potato cells swell and the length of the cylinder increases. (1)

(b) This is because the water potential of the potato cells and the sucrose solution become equal after 40 minutes. (1)
There is no net movement of water molecules into or out of the potato cells. (1)

(c) The water potential of the 0.4 M and 0.6 M sucrose solutions are lower than that of the cytoplasm of the potato cells, so the potato cylinders lose water by osmosis. (1)
The difference in water potential between the potato cells and the 0.6 M sucrose solution is greater than that between the potato cells and the 0.4 M sucrose solution. (1)
Therefore, the rate of osmosis is greater between the potato cells and the 0.6 M sucrose solution (1)
and hence the rate of decrease in the length in the potato cylinder in 0.6 M sucrose solution is higher. (1)

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(d) It provides a means of support in the young seedlings. (1)

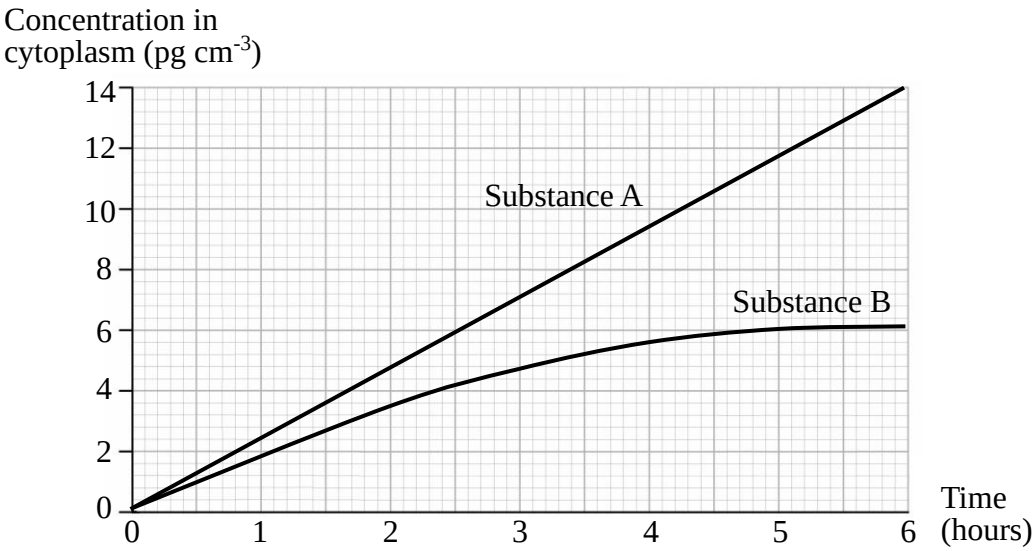
(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	3
(b)	Applying concepts	2
(c)	Applying concepts	4
(d)	Understanding basic concepts	1

|!|BCELB0040000008|!| ★

An experiment was carried out to study the uptake of substances A and B by *amoeba*. It is known that amoeba takes up these two substances by different mechanisms. *Amoebae* were placed in a solution with equal concentrations of substances A and B. The experimental set-up was kept at 15 ºC. The concentrations of substances A and B in the cytoplasm of the *amoeba* were measured at 30-minute intervals over a period of six hours. The following graph shows the results.



Note: 1 pg = 10^{-12} g

(a) Compare the uptake rate of substances A and B by the *amoeba* during the six hours. (4 marks)

(b) According to the graph, state and explain the mechanism by which *amoeba* takes up substance B. (4 marks)

(c) If the temperature is increased to 25 ºC, what would you expect the uptake rate of substance B to change? Explain your answer. (3 marks)

(d) By what mechanism does *amoeba* take up substance A? (1 mark)

##

(a) Substance A was taken up by the *amoeba* at a constant rate throughout the six hours. (1)
Substance B was taken up during the first five hours (1)
and the uptake stopped after five hours. (1)
The uptake rate of substance A was greater than that of substance B. (1)

(b) by diffusion (1)
The uptake rate was higher in the beginning of the experiment as the concentration difference between the *amoeba* and the solution was larger. (1)
The uptake rate decreased as the concentration difference between the *amoeba* and the solution became smaller. (1)
The uptake finally stopped when the concentration of substance B inside the *amoeba* was the same as that outside the *amoeba*. (1)

(c) The uptake rate of substance B will increase. (1)
This is because an increase in the temperature will increase the kinetic energy of molecules. (1)
Therefore, the molecules of substance B will move into the *amoeba* faster. (1)

(d) active transport (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4

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(b)	Analysing data	1
	Applying concepts	3
(c)	Applying concepts	3
(d)	Analysing data	1

|!|BCELB0040000009|!| ★

The water potentials of four plant root cells are shown in the table below. (kPa is a unit of pressure.)

Cell	Water potential (kPa)
A	–1800
B	–2100
C	–530
D	–200

The cells were placed in sucrose solution with a water potential of –530 kPa.

(a) Which cell(s) would have a net uptake of water? (1 mark)

(b) Which cell would initially take in water at the highest rate? Explain your answer. (3 marks)

(c) Describe and explain the appearance of cell D after it is placed in the sucrose solution for several hours. (4 marks)

(d) Explain why there was no change in the size of cell C. (2 marks)

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##

(a) cell A and cell B (1)

(b) cell B (1)

The rate of osmosis and hence the rate of water uptake depends on the water potential gradient. (1)

The difference between the water potential of cell B and the sucrose solution was the largest. (1)

Cell B therefore took in water at the highest rate initially.

(c) The water potential of cell D was higher than that of the sucrose solution. (1)

There was a net movement of water molecules from cell D to the sucrose solution. (1)

The cytoplasm shrank and detached from the cell wall. (1)

Cell D became plasmolysed. (1)

(d) The water potential of cell C was the same as that of the sucrose solution. (1)

There was no net water movement between cell C and the sucrose solution. (1)

(10 marks)

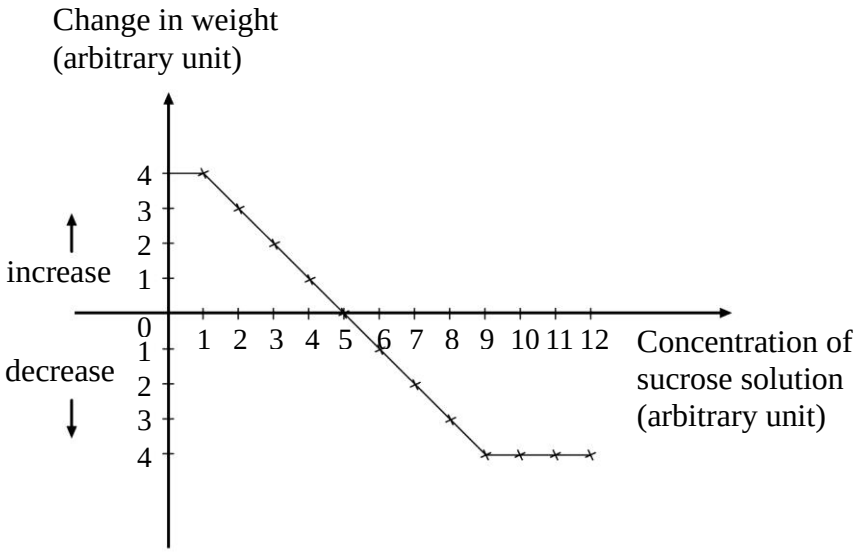
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	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
	Applying concepts	2
(c)	Analysing data	1
	Applying concepts	3
(d)	Analysing data	1
	Applying concepts	1

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|!|BCELB0040000010|!| ★ **SI**

The following graph shows the results of an experiment in which potato strips were immersed in sucrose solutions of various concentrations.



(a) How should we treat the potato strips before weighing them? (1 mark)

(b) Explain the change in the weight of the potato strips immersed in sucrose solutions of the following concentrations.

(i) 3 arbitrary units (2 marks)

(ii) 5 arbitrary units (2 marks)

(iii) 7 arbitrary units (2 marks)

(c) Explain why the following precautions should be taken.

(i) Potato strips with equal size were used. (1 mark)

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(ii) Sucrose solutions and potato strips were put in closed containers. (1 mark)

(d) Besides the weight, state another parameter that can be measured in this experiment. (1 mark)

(e) What is the state of potato cells in the potato strip which is immersed in sucrose solution of a concentration of 2 arbitrary units? (1 mark)

##

(a) Blot the excess solution on the surface of the potato strips with tissue paper. (1)

(b) (i) Since the potato strips were immersed in a hypotonic solution, (1)
water enters the cells by osmosis. (1)
This leads to an increase in the weight of the potato strips.

(ii) Since the potato strips were immersed in an isotonic solution, (1)
there is no net movement of water molecules. (1)
There is no change in the weight of the potato strips.

(iii) Since the potato strips were immersed in a hypertonic solution, (1)
water leaves the cells by osmosis. (1)
This leads to a decrease in the weight of the potato strips.

(c) (i) This could lead to easy / distinct comparison among the potato strips. (1)

(ii) This could avoid water loss by evaporation, which will change the water potential of the solution. (1)

(d) the length of the potato strips (1)

(e) The cells become turgid. (1)

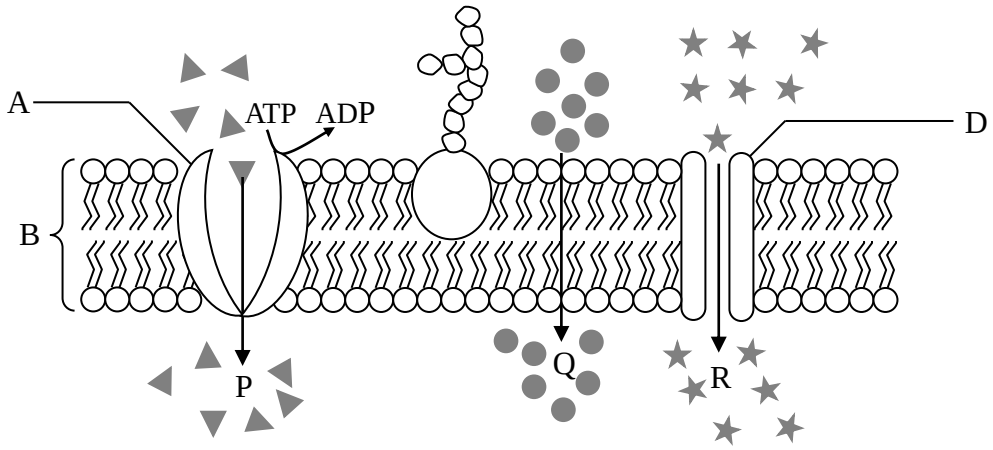
(11 marks)

	Skills assessed	Marks
(a)	Practical skills	1
(b)(i)	Analysing data	1
	Applying concepts	1
(b)(ii)	Analysing data	1
	Applying concepts	1
(b)(iii)	Analysing data	1
	Applying concepts	1
(c)(i)	Practical skills	1
(c)(ii)	Practical skills	1
(d)	Applying concepts	1
(e)	Applying concepts	1

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|!|BCELB0040000011|!| ★

The diagram below shows the transport of different substances across the cell membrane.



(a) Name the transport processes labelled by P, Q and R. (3 marks)

(b) (i) How does the reaction involving ATP facilitate the transport process P? (3 marks)

(ii) How do structures B and D facilitate the transport processes Q and R respectively? (4 marks)

(c) Suggest what the triangular and round molecules are most likely to be. (2 marks)

##

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(a) P: active transport (1)
Q: (simple) diffusion (1)
R: (facilitated) diffusion (1)

(b) (i) A is a carrier protein. (1)
It changes shape with the utilization of energy / ATP. (1)
The molecules bound to A are then transported to the other side of the cell membrane. (1)

(ii) B is a phospholipid bilayer which is permeable to lipid-soluble substances. (1)
Lipid-soluble substances can dissolve in layer B and pass directly through the layer by diffusion. (1)
D is a channel protein. (1)
Molecules can pass through the membrane by passing through the pore formed by D. (1)

(c) Triangular molecules: ions / glucose / amino acid / other reasonable answer (1)
Round molecules: lipid / vitamin A / vitamin D / other reasonable answer (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)(i)	Analysing data	1
	Applying concepts	2
(b)(ii)	Analysing data	2
	Applying concepts	2
(c)	Applying concepts	2

|!|BCELB0040000012|!| ★

Read the passage below and answer the questions that follow.

Water intoxication

An American woman joined a water-drinking contest. However, she found her head painful and was later found dead in her house. Coroner inquest had shown that her death was related to water intoxication.

Water is essential to our bodies, but drinking too much water will cause the brain to swell. This prevents the brain from regulating vital functions and is lethal. However, our bodies can regulate the water content in the body and the water potential of the blood. If we drink too much water, the kidneys will be unable to work fast enough to excrete the excess water, so the blood becomes more dilute.

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(a) Explain why drinking too much water will cause the brain to swell. (3 marks)

(b) Besides the swelling of the brain, explain why drinking too much water is lethal. (2 marks)

(c) (i) List ***two*** ways that can regulate the water content in the body. (2 marks)

(ii) How can the above ways regulate the water potential of the blood? (2 marks)

##

(a) Drinking too much water will dilute the body fluid. (1)
This causes the body fluid to be more hypotonic / have a higher water potential than the brain cells. (1)
Water then enters the brain cells by osmosis and makes the brain cells swell. (1)

(b) Drinking too much water will dilute the blood. (1)
This will lead to a net movement of water molecules from the blood to the body cells and cause the cells to burst. (1)

(c) (i) urination (1)
sweating (1)

(ii) Urination and sweating can remove excess water from the body. (1)
The blood will become less dilute and its water potential decreases / returns to normal level. (1)

(9 marks)

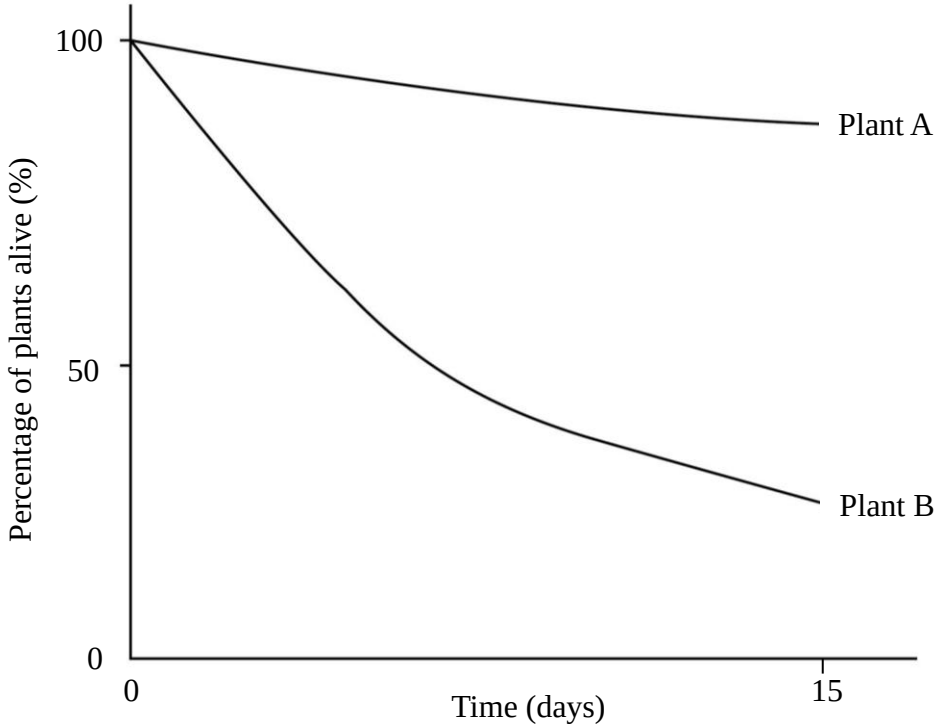
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Applying concepts	2

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|!|BCELA0040000013|!| ★★ **Cross topic**

Nancy performed an experiment to find out the effect of a solution of high salt concentration on the growth of two groups of plants, A and B. She poured the salt solution to the pots daily for 15 days.



(a) Explain why most of the leafy shoots of plant B withers. (3 marks)

(b) Give a term to describe the salt solution in relation to its water potential. (1 mark)

(c) Draw a labelled diagram to show a root cell from plant B. (4 marks)

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(d) Plant A lives in an adverse habitat with high salinity.

(i) Suggest a habitat in which plant A lives. (1 mark)

_____

(ii) Suggest an adaptation of plant A so that it can survive in high salt environment. (1 mark)

_____

##

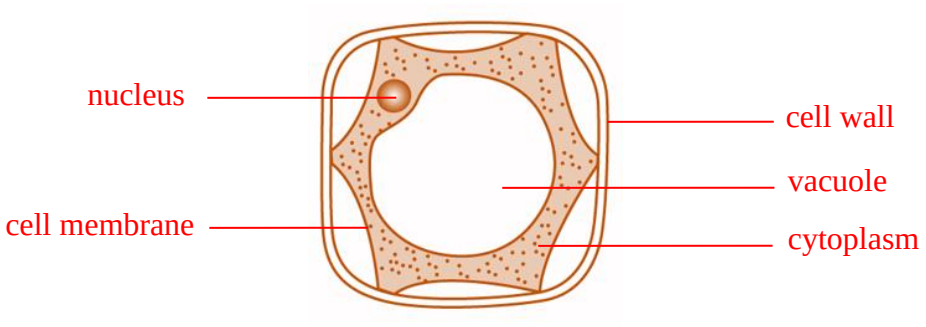
(a) The salt solution has a lower water potential than that of the root cells. (1)

Water moves out of the root cells by osmosis. (1)

The flaccid plant cells do not have enough water to support the plant body. (1)

(b) Hypertonic (1)

(c)



A plasmolysed root cell from plant B

Correct drawing (1)

Correct labels (Any ***two***) (2)

Correct title (1)

(d) (i) mangrove / other reasonable answer (1)

(ii) The leaf of Plant A has salt glands to remove excess salt from the body. (1)

(10 marks)

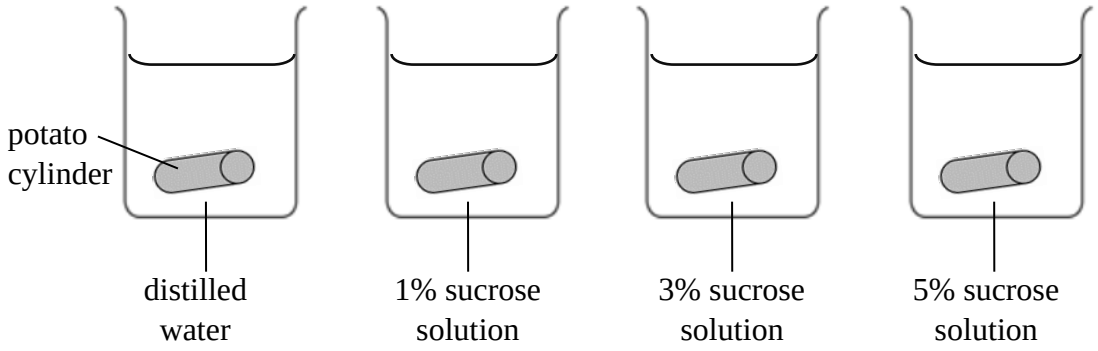
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Understanding basic concepts	1
(c)	Presenting information	4
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	1

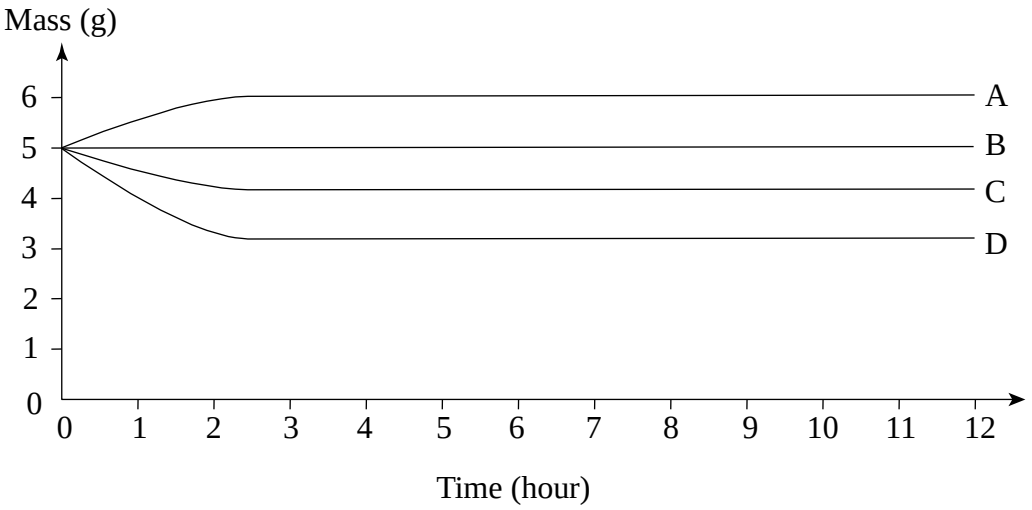
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|!|BCELB0040000014|!| ★ **SI**

Four potato cylinders with equal length and mass were freshly prepared from a peeled potato. They were placed in four beakers of different liquids as shown below.



Their masses were recorded and presented graphically as shown below.



(a) What is the possible objective of this experiment? (1 mark)

_____

_____

(b) Which process does this experiment demonstrate? What molecule(s) is/are involved in this process? (2 marks)

_____

_____

(c) What is the independent variable in this experiment? (1 mark)

_____

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(d) Describe and explain the shape of curve D. (4 marks)

(e) Another identical potato cylinder was put into a 10% sucrose solution (not shown in the diagram). Sketch a graph to show the results for this potato cylinder. (3 marks)

##

(a) To study how the sucrose concentration affects the mass of potato cylinders / other reasonable answer (1)

(b) osmosis (1)
water molecules (1)

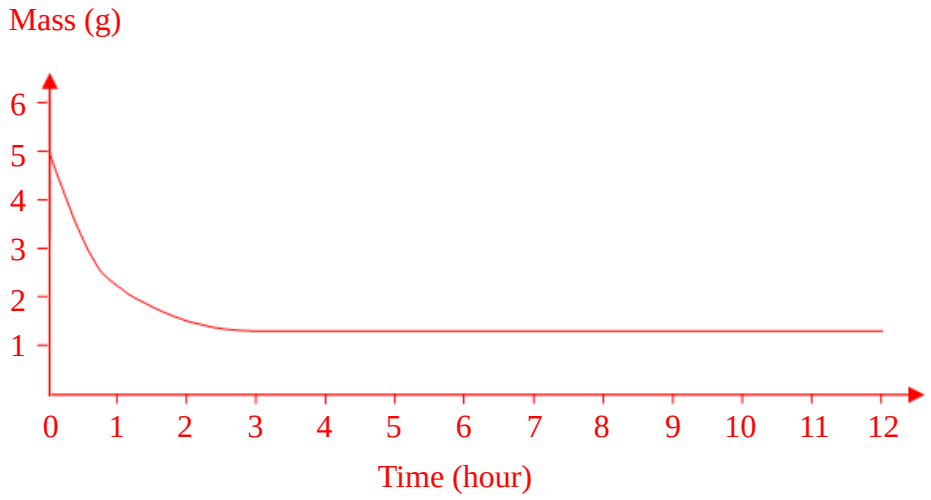
(c) the concentration of sucrose solution (1)

(d) The water potential of the sucrose solution is higher than that inside the potato cell at the start of the experiment. (1)
There is a net movement of water molecules out of the potato cells and this leads to a decrease in the mass of the potato cylinder. (1)
After three hours, the water potentials of the sucrose solution and the inside of the potato cylinders become the same. (1)
There is no net movement of water molecules and there is no change in mass of the potato cylinder. (1)

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(e)

Change in the mass of potato cylinder in 10% sucrose solution



Correct graph (1)
Correct labelling of axis (1)
Correct title (1)

(11 marks)

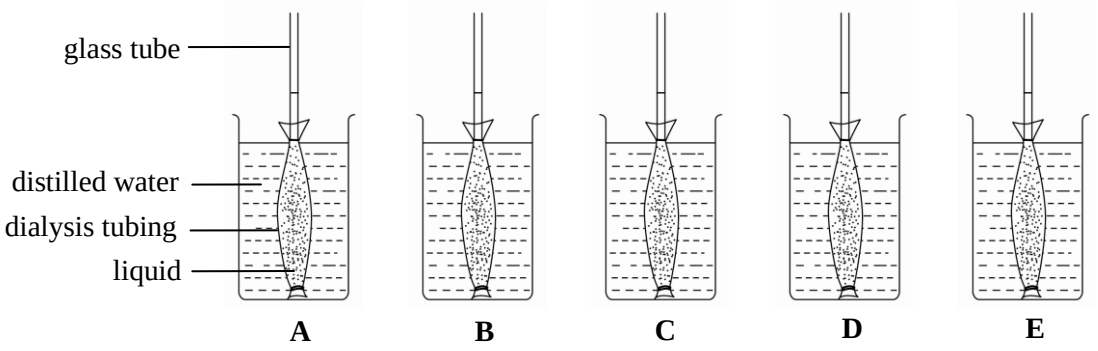
##

	Skills assessed	Marks
(a)	Practical skills	1
(b)	Analysing data	1
	Applying concepts	1
(c)	Practical skills	1
(d)	Analysing data	2
	Applying concepts	2
(e)	Presenting information	3

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|!|BCELA0040000015|!| ★★ **SI**

The diagram below shows an experiment on osmosis.

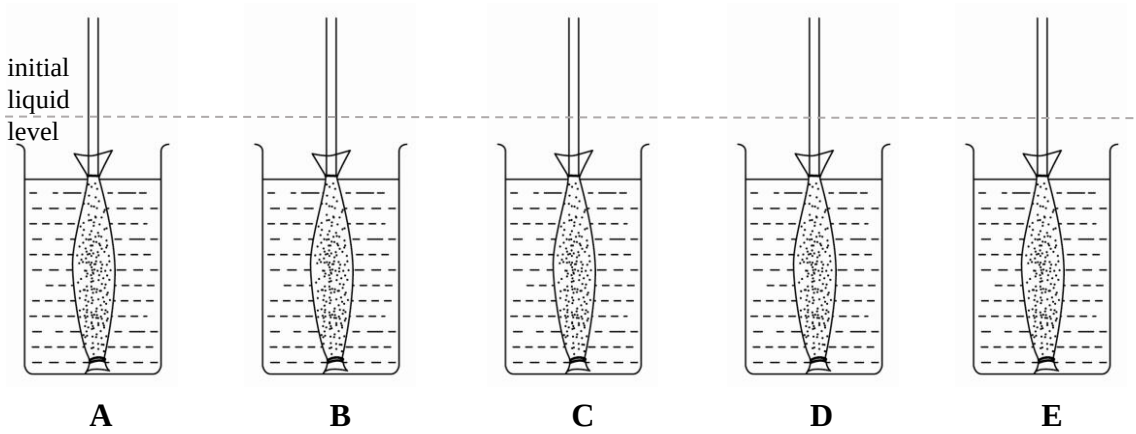


The following table shows the liquid inside the dialysis tubing in each set-up.

Dialysis tubing	Liquid inside the dialysis tubing
A	distilled water
B	5% glucose solution
C	5% sucrose solution
D	10% glucose solution
E	10% sucrose solution

The liquid level in each glass tube was the same at the beginning. The set-ups were left undisturbed for 30 minutes.

(a) Mark on the diagrams below the expected liquid level in the glass tube of each set-up after 30 minutes. (2 marks)



(b) Why are glass tubes with a narrow bores more preferred in this experiment?

(1 mark)

(c) After the dialysis tubings were filled with the liquids, they were put under a stream of running tap water for a while. Explain why this step is necessary. (2 marks)

(d) Compare the results of set-ups B and C. Explain the difference(s). (7 marks)

##

(a) A, B and D: the liquid levels remain unchanged (1)
C and E: the liquid levels are higher than the initial liquid level and the liquid level in E is higher than that in C. (1)

(b) A glass tube with a narrower bore enables more obvious changes in liquid level. (1)

(c) To wash away any liquid left on the outer surface of the dialysis tubing. (1)
The liquid left could affect the water potential of the distilled water outside and affect the rate of osmosis. (1)

(d) The liquid level of the glass tube in set-up C is higher than that in set-up B. (1)
This is because in set-up B, glucose molecules can pass through the dialysis tubing and diffuse from the dialysis tubing to the distilled water. (1)
The glucose concentrations of the liquid inside and outside the dialysis tubing are equal finally (1)
and there is no net movement of water molecules between the inside and outside of the dialysis tubing. (1)
Therefore, there is no change in the liquid level in set-up B.
In set-up C, as the sucrose molecules are too large to leave the dialysis tubing, (1)
the water potential outside the dialysis tubing is higher than that inside. (1)
Water enters the dialysis tubing by osmosis and the liquid level rises. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Practical skills	1

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(c)	Practical skills	2
(d)	Analysing data	2
	Applying concepts	5

|!|BCELB0040000016|!| ★ **SI**

A student carries out an experiment to demonstrate a certain transport process in the living cells of a carrot.

A large carrot was cut into many small discs of similar size. The initial weight of every piece of carrot was measured and recorded. Five pieces of carrot were separately added to five different test tubes containing sucrose solution of different concentrations. After four hours, the weight of the carrots was measured again. The results are shown in the table below.

Test tube	A	B	C	D	E
Concentration of sucrose solution (M)	0.00	0.10	0.20	0.30	0.40
Initial weight (g)	21.4	23.3	19.6	19.9	22.1
Final weight (g)	21.9	23.5	19.5	19.5	21.6
The percentage change in weight (%)		+0.85		–2.01	

(a) Complete the table above. (1 mark)

(b) On the graph paper provided, plot a graph to show the results. (4 marks)

(c) From the graph, which concentration of sucrose solution has the same water potential as the carrot cells? (1 mark)

_____

(d) Some cells were then taken from the carrot discs in test tubes C and E and put onto a slide for observation under a microscope.

(i) State ***two*** observable differences between the cells in test tubes C and E. (2 marks)

_____

_____

_____

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(ii) Name the process that causes the differences. (1 mark)

(e) If animal cells were put into solution A, what would happen to the cells? (2 marks)

(f) Suggest an improvement when performing this experiment in order to yield more reliable results. (1 marks)

##

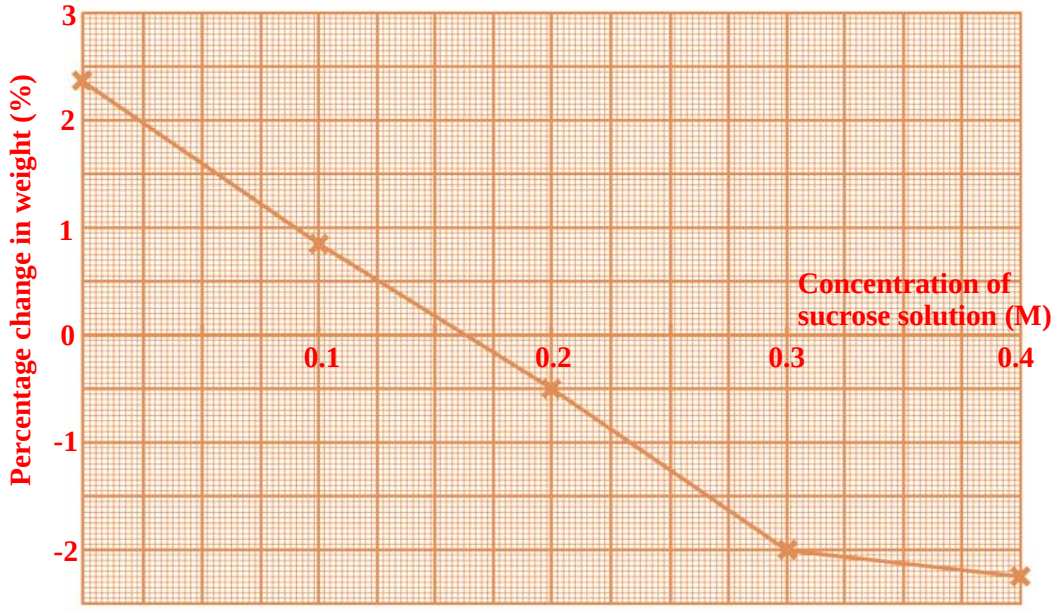
(a)

Test tube	A	B	C	D	E
Percentage change in weight (%)	**+2.34**	+0.85	**−0.51**	–2.01	**−2.26**

(+ / – should be included) (1)

(b)

Graph showing the percentage change in weight of carrot in sucrose solutions of different concentrations



X- and Y-axes with correct names, units and scales (2)

Correct plotting and joining of the five points (1)

Correct title (1)

(c) Accept answers within the range of 0.15 to 0.18 M. (1)

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(d) (i) The cell membrane is detached from the cell wall in the cells from test tube E but remains attached to the cell wall in the cells from test tube C. (1)
The cytoplasm / vacuole shrinks in the cells from test tube E but not in those from test tube C. (1)

(ii) osmosis (1)

(e) The animal cells would swell (1)
and eventually burst. (1)

(f) Add several carrot discs instead of one into each test tube and obtain an average value of the change in the weight. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	1
(b)	Presenting information	4
(c)	Analysing data	1
(d)(i)	Applying concepts	2
(d)(ii)	Understanding basic concepts	1
(e)	Applying concepts	2
(f)	Practical skills	1

|!|BCELC0040000017|!|

(a) Explain why the cell membrane is described as 'fluid mosaic'. (2 marks)

(b) Draw a labelled diagram to show a phospholipid bilayer. (2 marks)

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(c) What are the properties of the phosphate group 'head' and the fatty acid 'tail' of the phospholipid molecule in relation to water? (2 marks)

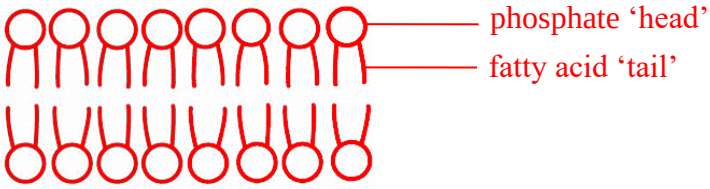
(d) State ***two*** functions of the 'mosaics' in the cell membrane. (2 marks)

##

(a) Fluid: the phospholipid molecules and some protein molecules can move laterally. (1)

Mosaic: the protein molecules are interspersed among the phospholipid molecules in a mosaic pattern. (1)

(b)



The phospholipid bilayer

Drawing the head and tail correctly (1)

Heads are on the outer and inner surfaces of the membrane, the tails are in the middle. (1)

(c) The phosphate group 'head' is hydrophilic. (1)

The fatty acid 'tail' is hydrophobic. (1)

(d) Any ***two*** of the following: (2)

- receptor proteins bind with specific molecules which can then trigger or inhibit certain cell activities
- enzymes control chemical reactions
- channel proteins form pores through which water molecules, ions and small water soluble molecules can pass
- carrier proteins are involved in active transport
- glycoproteins may act as recognition markers

(8 marks)

##

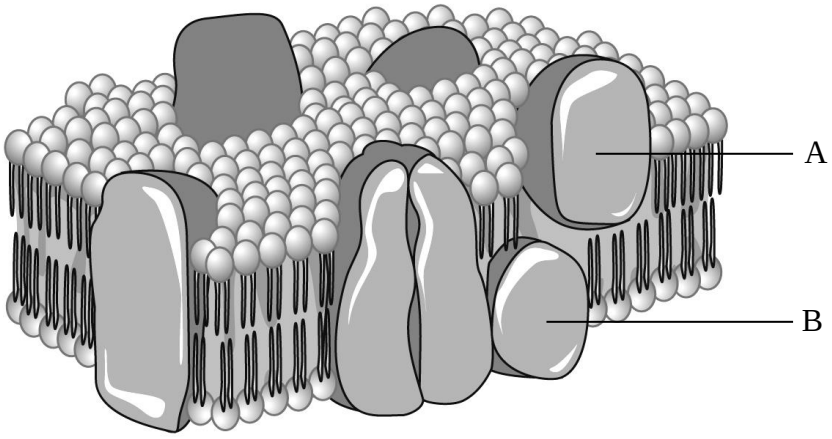
	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Presenting information	2

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(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

|!|BCELB0040000018|!| ★ **Cross topic**

The diagram below shows a part of the cell membrane. Structure A is facing the outside of the cell while structure B is facing the inside of the cell.



(a) (i) If this is the cell membrane of a bacterium, what may structure A be? (1 mark)

_____

(ii) If this is the cell membrane of a human white blood cell, what may structure A be? (1 mark)

_____

(iii) What happens when the structure A of a bacterial cell combines with the structure A of a human white blood cell? (2 marks)

_____

_____

_____

(b) In the extraction of chlorophyll from a green leaf, the leaf should be immersed in warm liquid X.

(i) What is liquid X? (1 mark)

_____

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(ii) Explain why chlorophyll can be extracted with liquid X. Explain with reference to the property of the cell membrane. (3 marks)

##

(a) (i) antigen (1)

(ii) receptor for antigen (1)

(iii) After bacterial antigen has combined with the receptor, antigens are recognized by the white blood cell. (1)

This will stimulate immune response against the bacterium. (1)

(b) (i) alcohol (1)

(ii) The cell membrane is made up of phospholipids. (1)

Organic solvent such as alcohol dissolves the phospholipids. (1)

Both the cell membrane and the membrane of the chloroplast break and chlorophyll is released. (1)

(8 marks)

##

	Skills assessed	Marks
(a)(i)	Applying concepts	1
(a)(ii)	Applying concepts	1
(a)(iii)	Applying concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Applying concepts	3

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